

Homework 5

Ans 1 (a)  $\sin \alpha = \frac{b}{R} \Rightarrow b = R \sin \alpha$

$$\theta + 2\alpha = \pi \Rightarrow \alpha = \frac{\pi - \theta}{2}$$

$$\Rightarrow b = R \sin \{(\pi - \theta)/2\}$$

$$= R \cos(\theta/2)$$

(b)  $D(\theta) \equiv d\sigma/d\Omega$

$$d\sigma = b db d\phi$$

$$d\Omega = \sin \theta d\theta d\phi$$

$$D(\theta) = \frac{b}{\sin \theta} \left| \frac{db}{d\theta} \right|$$

now  $\frac{db}{d\theta} = -\frac{R}{2} \sin(\theta/2)$

$$\Rightarrow D(\theta) = \cancel{R} \frac{R \cos(\theta/2)}{\sin \theta} \frac{R}{2} \sin(\theta/2)$$

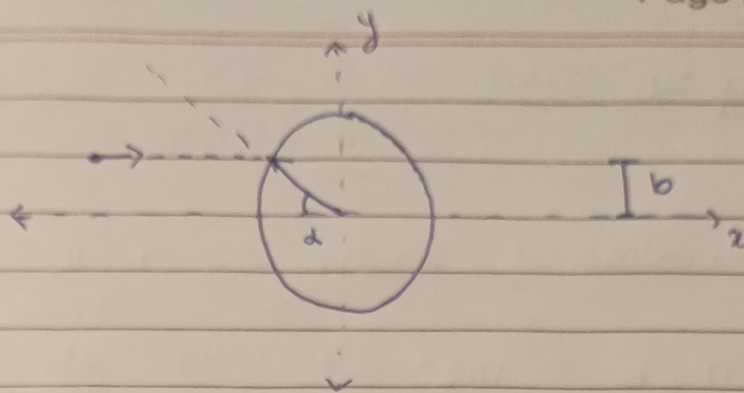
$$\Rightarrow \boxed{D(\theta) = R^2/4}$$

(c)  $\sigma_T = \int D(\theta) d\Omega = \int_0^{2\pi} \int_0^\pi D(\theta) \sin \theta d\theta d\phi$

$$D(\theta) = R^2/4 : \quad \sigma_T = \frac{R^2}{4} \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta$$

$$\Rightarrow \sigma_T = \frac{R^2 \pi}{2} \int_0^\pi \sin \theta d\theta = \pi R^2$$

Ans 2 - (a)



$$b = R \sin \alpha$$

$$\phi + 2\alpha = \pi \Rightarrow \alpha = \frac{\pi - \phi}{2}$$

$$\Rightarrow b = R \cos(\phi/2)$$

$$(b) \quad D(\phi) \equiv |dp| d\phi = |-(R/2) \sin(\phi/2)| \\ = \frac{R}{2} \sin\left(\frac{\phi}{2}\right)$$

$$(c) \quad \sigma_T = \int_0^{2\pi} D(\phi) d\phi = \frac{R}{2} \int_0^{2\pi} \sin(\phi/2) d\phi \\ = 2R$$

Ans 3 - (a)  $|p_i| = |p_f| = |p| = mv_0$ 

$$|\Delta p| = |p_f - p_i| = 2p \sin(\theta/2)$$

According to Newton's second law, change in momentum equals to the impulse delivered by Coulomb force

$$F = (4\pi\epsilon_0)^{-1} (q_1 q_2 / r^2)$$

$$|\Delta p| = \int F_{\text{sym}} dt$$

where  $F_{\text{sym}}$  is the component of the Coulomb force along the axis of symmetry.  $F_{\text{sym}} = F \cos \psi$

$$\Rightarrow 2p \sin(\theta/2) = \int_{-\infty}^{\infty} \frac{q_1 q_2}{4\pi\epsilon_0 r^2} \cos\psi \, dt$$

by conservation of angular momentum  $L$ :

$$L = mv_\infty b = m r^2 \frac{d\psi}{dt} \Rightarrow \frac{dt}{r^2} = \frac{m}{L} d\psi$$

now, by impulse equation:

$$2p \sin(\theta/2) = \int_{-\psi_{\max}}^{\psi_{\max}} k \cos\psi \left(\frac{m}{L}\right) d\psi = \frac{km}{L} \int_{-\psi_{\max}}^{\psi_{\max}} \cos\psi \, d\psi$$

where  $k = q_1 q_2 / 4\pi\epsilon_0$ .

and  $\psi_{\min}, \psi_{\max} = \pm \{(\pi - \theta)/2\}$

$$\Rightarrow 2mv_\infty \sin(\theta/2) = \frac{2km}{mv_\infty b} \cos\left(\frac{\theta}{2}\right)$$

$$\Rightarrow v_\infty^2 \sin(\theta/2) = \frac{k}{mb} \cos\left(\frac{\theta}{2}\right)$$

$$E = \frac{1}{2} m v_\infty^2 \Rightarrow m v_\infty^2 = 2E$$

$$\Rightarrow \boxed{b = \frac{q_1 q_2}{8\pi\epsilon_0 E} \cot\left(\frac{\theta}{2}\right)}$$

$$(b) \quad D(\theta) = \frac{b}{\sin \theta} \left| \frac{db}{d\theta} \right|$$

$$\frac{db}{d\theta} = \frac{-q_1 q_2}{16 \pi \epsilon_0 E \sin^2(\theta/2)}$$

$$\Rightarrow D(\theta) = \frac{q_1 q_2^2 \cos(\theta/2) \sin^{-2}(\theta/2)}{2 \sin(\theta/2) \cos(\theta/2) \sin(\theta/2) 8(\pi \epsilon_0 E)^2 16}$$

$$\Rightarrow D(\theta) = \left[ \frac{q_1 q_2}{16 \pi \epsilon_0 E \sin^2(\theta/2)} \right]^2$$

$$(c) \quad \sigma_T = \int D(\theta) d\Omega = \int_0^{2\pi} d\phi \int_0^\pi D(\theta) \sin \theta d\theta$$

$$\Rightarrow \sigma_T = 2\pi \left( \frac{q_1 q_2}{16 \pi \epsilon_0 E} \right)^2 \int_0^\pi \frac{\sin \theta}{\sin^2(\theta/2)} d\theta$$

$$\sin \theta = 2 \sin(\theta/2) \cos(\theta/2)$$

$$\Rightarrow \sigma_T = 4\pi \left( \frac{q_1 q_2}{16 \pi \epsilon_0 E} \right)^2 \int_0^\pi \frac{\cos(\theta/2)}{\sin^3(\theta/2)} d\theta$$

$\sigma_T$  diverges at  $\theta \rightarrow 0$ . This happens because Coulomb potential  $V(r) \propto 1/r$  has infinite range.

Ans 4- (a)  $\psi(r) = \sum_{l=0}^{\infty} i^l (2l+1) P_l(\cos \theta) [a_l j_l(kr) + b_l n_l(kr)]$

$$j_l(kr) \sim \frac{\sin(kr - l\pi/2)}{kr} = \frac{e^{i(kr - l\pi/2)} - e^{-i(kr - l\pi/2)}}{2ikr}$$

$$n_l(kr) \sim \frac{-\cos(kr - l\pi/2)}{kr} = -\frac{e^{i(kr - l\pi/2)} + e^{-i(kr - l\pi/2)}}{2kr}$$

$$\Rightarrow j_l \sim \frac{e^{ikr} i^{-l} - e^{-ikr} i^l}{2ikr}$$

$$n_l \sim \frac{e^{ikr} i^{-l} + e^{-ikr} i^l}{2kr} = \frac{e^{ikr} i^{-l} + e^{-ikr} i^l}{2ikr \cdot i}$$

substituting into  $\psi(\vec{r})$ :

$$\psi(\vec{r}) \sim \sum_{l=0}^{\infty} \frac{(2l+1)}{2i} P_l(\cos \theta) \left[ (a_l - ib_l) \frac{e^{ikr}}{kr} + (-1)^{l+1} (a_l + ib_l) \frac{e^{-ikr}}{kr} \right]$$

(b)  $\psi(R) = 0 \Rightarrow$  that phase shift of  $\delta_l$  will occur.

$$a_l - ib_l = e^{2i\delta_l} \quad a_l + ib_l = 1$$

$$\Rightarrow 2a_l = e^{2i\delta_l} + 1 = 2e^{i\delta_l} \cos \delta_l$$

$$\Rightarrow a_l = e^{i\delta_l} \cos \delta_l$$

$$\Rightarrow \psi(\vec{r}) = \sum_{l=0}^{\infty} i^l (2l+1) P_l(\cos \theta) e^{i\delta_l} [\cos \delta_l j_l(kr) - \sin \delta_l n_l(kr)]$$

(c)  $\psi(R) = 0$

$$\cos \delta_l j_l(kR) - \sin \delta_l n_l(kR) = 0$$

$$\Rightarrow \sin \delta_l j_l(kR) = (\cos \delta_l) n_l(kR)$$

$$\Rightarrow \boxed{\tan \delta_l = \frac{j_l(kR)}{n_l(kR)}}$$

(d)  $l=0$  :  $j_0(x) = \frac{\sin x}{x}$        $n_0(x) = -\frac{\cos x}{x}$

$$\Rightarrow \tan \delta_0 = -\tan(kR) \Rightarrow \delta_0 = -kR$$

$$l=1 : j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x}$$

$$n_1(x) = -\frac{\cos x}{x^2} - \frac{\sin x}{x}$$

$$l=2 : j_2(x) = \left( \frac{3}{x^3} - \frac{1}{x} \right) \sin x - \frac{3}{x^2} \cos x$$

$$n_2(x) = -\left( \frac{3}{x^3} - \frac{1}{x} \right) \cos x - \frac{3}{x^2} \sin x$$

| $l$ | $\tan \delta_l$ for $kR = \pi/6$ | $\tan \delta_l$ for $kR = \pi/4$ |
|-----|----------------------------------|----------------------------------|
| 0   | $-\pi/6$                         | $-\pi/4$                         |
| 1   | $-0.0417$                        | $-0.1202$                        |
| 2   | $-0.0068$                        | $-0.0057$                        |

(e)

$$kR \ll 1$$

$$\tan \delta_l = \frac{j_l(kR)}{n_l(kR)} \approx \left\{ \frac{(kR)^l}{(2l+1)!!} \right\} \left\{ \frac{-(2l-1)!!}{(kR)^{2l+1}} \right\}$$

$$\Rightarrow \tan \delta_l \approx \frac{-(kR)^{2l+1}}{(2l+1)!! (2l-1)!!}$$

$$\text{for } kR \ll 1: \quad \delta_l \approx \frac{-(kR)^{2l+1}}{(2l+1)!! (2l-1)!!}$$

(f)

$$\sigma_T = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l$$

(i) For  $kR = \pi/6$ :

$$\sum_{l=0}^2 (2l+1) \sin^2 \delta_l \approx 0.2551$$

$$\sigma_T = \frac{4\pi (0.2551)}{k^2} \approx 3.72 \pi R^2$$

(ii)  $kR = \pi/4$ 

$$\sum_{l=0}^2 (2l+1) \sin^2 \delta_l \approx 0.5429$$

$$\sigma_T = \frac{4\pi (0.5429)}{k^2} \approx 3.52 \pi R^2$$

$$(g) \quad \sin^2 \delta_0 = \sin^2(KR) \approx \frac{(KR)^2 - (KR)^4}{3}$$

$$\delta_1 \approx \tan \delta_1 \approx -\frac{(KR)^3}{3}$$

$$\sin^2 \delta_1 \approx \left(\frac{-\cancel{x^3}}{3}\right)^2 = \frac{\cancel{x^6}}{9} = \frac{(KR)^6}{9}$$

$$\sigma_T = \frac{4\pi}{k^2} [\sin^2 \delta_0 + 3\sin^2 \delta_1 + \dots]$$

$$= \frac{4\pi}{k^2} [(KR)^2 - \frac{1}{3}(KR)^4 + O(KR)^6]$$

$$= 4\pi R^2 \left[ 1 - \frac{1}{3}(KR)^2 + O(KR)^4 \dots \right]$$

$$\Rightarrow \boxed{\sigma_T = 4\pi R^2 \left[ 1 - \frac{(KR)^2}{3} + \dots \right]}$$

Ans 5-  $\sigma_T = 4\pi R^2 \left[ 1 - \frac{\tan(qR)}{qR} \right]^2$

$$\frac{-\hbar^2}{2m} \frac{d^2 u(r)}{dr^2} + V(r)u(r) = E u(r)$$

potential well:  $-V_0, \quad r \leq R$   
 $0, \quad r > R$

inside:  $\frac{d^2 u_1}{dr^2} + q^2 u_1 = 0, \quad q^2 = \frac{2m(E+V_0)}{\hbar^2}$

$$u_{\text{in}}(r) = A \sin(qr) + B \cos(qr)$$

$$\psi(r) = u(r)/r \quad \text{must be finite at } r=0$$

$$\Rightarrow u_{\text{in}}(0) = 0$$

$$\Rightarrow B = 0$$

$$\Rightarrow u_{\text{in}}(r) = U \sin(qr)$$

$$\text{for } r > R: \frac{d^2 u_{\text{out}}}{dr^2} + k^2 u_{\text{out}} = 0, \quad k^2 = \frac{2mE}{\hbar^2}$$

$$\Rightarrow u_{\text{out}}(r) = C \sin(kr + \delta_0)$$

$$\text{Boundary cond: } u_{\text{out}}(R) = u_{\text{in}}(R)$$

$$u'_{\text{out}}(R) = u'_{\text{in}}(R)$$

$$\Rightarrow \frac{qA \cos(qR)}{A \sin(qR)} = \frac{kC \sin(kR + \delta_0)}{C \sin(kR + \delta_0)}$$

$$\Rightarrow q \cot(qR) = k \cot(kR + \delta_0)$$

in low energy limit,  $k \rightarrow 0 \Rightarrow \delta_0 \rightarrow \text{small}$

$$k \cot(kR + \delta_0) = \frac{k \cos(kR + \delta_0)}{\sin(kR + \delta_0)}$$

$$\approx \frac{k}{kR + \delta_0}$$

$$\Rightarrow q \cot(qR) = \frac{k}{kR + \delta_0}$$

$$\Rightarrow S_0 = k \left( \frac{\tan(qR)}{q} - R \right)$$

$$\sigma_T = \frac{4\pi}{k^2} \sin^2(S_0), \quad S_0 \text{ is small}$$

$$\Rightarrow \sigma_T = 4\pi \left( R - \frac{\tan(qR)}{q} \right)^2$$

$$\Rightarrow \sigma_T = 4\pi R^2 \left[ 1 - \frac{\tan(qR)}{qR} \right]^2$$

Ans 6- (a) for  $l=0$ :  $-\frac{\hbar^2}{2m} u''(r) + V_0 \delta(r-R)u(r) = Eu(r)$

$$k^2 = 2mE/\hbar^2 : u''(r) + k^2 u(r) = \frac{2mV_0}{\hbar^2} \delta(r-R)u(r)$$

$$r < R : u(r) = A \sin kr$$

$$r > R : u(r) = B \sin(kr + S_0)$$

$$u_L(R) = u_R(R) \Rightarrow A \sin(kR) = B \sin(kR + S_0)$$

discontinuity in 1<sup>st</sup> derivative:

$$u'_>(R) - u'_<(R) = \frac{2mV_0}{\hbar^2} u(R)$$

$$P = 2mV_0 R / \hbar$$

$$\Rightarrow k B \cos(kR + S_0) - k A \cos(kR) = \frac{P}{R} A \sin(kR)$$

$$k \cot(kR + \delta_0) - k \cot(kR) = \beta / R$$

$$\text{let } x = kR, \Rightarrow \cot(x + \delta_0) = \frac{\beta}{x} + \cot x$$

$$\Rightarrow \frac{-\sin \delta_0}{(\sin x) \sin(x + \delta_0)} = \frac{\beta}{x}$$

$$\Rightarrow -x \sin \delta_0 = \beta \sin x \sin(x + \delta_0)$$

$$\Rightarrow \cancel{\cot} -x = \beta \sin x [\cos x + \cot \delta_0 \sin x]$$

$$\Rightarrow \cot \delta_0 = - \frac{(x + \frac{\beta}{2} \sin(2x))}{\beta \sin^2 x}$$

$$\text{now } \sin^2(\delta_0) = \frac{1}{1 + \cot^2(\delta_0)}$$

$$\Rightarrow \cot^2(\delta_0) = \frac{\left\{ x + \frac{\beta}{2} \sin(2x) \right\}^2}{\beta^2 \sin^4 x}$$

$$\Rightarrow 1 + \cot^2 \delta_0 = \frac{\beta^2 \sin^4 x + x^2 + x\beta \sin 2x + \frac{\beta^2}{4} \sin^2 2x}{\beta^2 \sin^4 x}$$

$$\begin{aligned} \beta^2 \sin^4 x + \beta^2 \sin^2 x \cos^2 x &= \beta^2 \sin^2 x \{ \sin^2 x + \cos^2 x \} \\ &= \beta^2 \sin^2 x \end{aligned}$$

$$\Rightarrow \text{Num} = x^2 + x\beta \sin(2x) + \beta^2 \sin^2 x$$

$$\Rightarrow \sin^2 \delta_0 = \frac{\beta^2 \sin^4 x}{x^2 + x\beta \sin 2x + \beta^2 \sin^2 x}$$

(b) let  $x = kR \ll 1 \Rightarrow \sin x \approx x$

$$\Rightarrow \sin^2 \delta_0 = \frac{\beta^2 x^4}{x^2 + x^2(2\beta) + \beta^2 x^2}$$

$$= \frac{\beta^2 x^2}{1 + \beta^2 + 2\beta} = \frac{\beta^2 x^2}{(1+\beta)^2}$$

$$\sigma_T = \frac{4\pi}{k^2} \sin^2 \delta_0 = \frac{4\pi}{k^2} \frac{\beta^2 k^2 R^2}{k^2 (1+\beta)^2}$$

$$\Rightarrow \boxed{\sigma_T = 4\pi R^2 \left[ \frac{\beta}{1+\beta} \right]^2}$$

Ans 7)  $q(\vec{r}, \vec{r}') = \int \frac{d^3 k'}{(2\pi)^3} \frac{e^{i\vec{k}' \cdot (\vec{r} - \vec{r}')}}{k^2 - (k')^2 \pm i\epsilon}$

$\vec{R} = \vec{r} - \vec{r}'$  and  $R = |\vec{R}|$

switch to spherical coordinates for  $k'$ , where polar angle  $\theta$  is measured relative to  $\vec{R}$

$$q(R) = \frac{1}{(2\pi)^3} \int_0^\infty (k')^2 dk' \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi \frac{e^{i\vec{k}' \cdot \vec{R} \cos\theta}}{k^2 - (k')^2 \pm i\epsilon}$$

$$= \frac{1}{(2\pi)^3} (2\pi) \int_0^\infty (k')^2 dk' \frac{2\sin(k'R)}{k'R} \frac{1}{k^2 - (k')^2 \pm i\epsilon}$$

$$= \frac{1}{2\pi^2 i R} \int_0^\infty \frac{k' \sin(k'R) dk'}{k^2 - (k')^2 \pm i\epsilon}$$

$$= \frac{1}{2\pi^2 i R} \int_{-\infty}^{\infty} \frac{k' e^{i k' R} dk'}{2i(k^2 - (k')^2 \pm i\epsilon)}$$

$$= \frac{-1}{4\pi^2 i} \int_{-\infty}^{\infty} \frac{k' e^{i k' R} dk'}{\{k' - (k \pm i\epsilon)\} \{k' + (k \pm i\epsilon)\}}$$

By Cauchy's residue theorem:

$$G(\vec{r}, \vec{r}') = \frac{-1}{4\pi} \frac{e^{\pm i k |\vec{r} - \vec{r}'|}}{|\vec{r} - \vec{r}'|}$$

Ans 8)  $\left[ \frac{-\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right] \psi(\vec{r}) = E \psi(\vec{r})$

$$E = \hbar^2 k^2 / 2m \Rightarrow (\nabla^2 + k^2) \psi(\vec{r}) = \frac{2m}{\hbar^2} V(\vec{r}) \psi(\vec{r})$$

general sol<sup>n</sup>:  $\psi(\vec{r}) = \psi_0(\vec{r}) + \int G(\vec{r}, \vec{r}') f(\vec{r}') d^3 r'$

the sol<sup>n</sup> to homogeneous eq<sup>n</sup> is  $\psi_0(\vec{r}) = e^{i \vec{k} \cdot \vec{r}}$

now substitute  $G(\vec{r}, \vec{r}')$  and  $f(\vec{r}')$ :

$$\psi(\vec{r}) = e^{i \vec{k} \cdot \vec{r}} + \int \left( \frac{-1}{4\pi} \frac{e^{i k |\vec{r} - \vec{r}'|}}{|\vec{r} - \vec{r}'|} \right) \left( \frac{2m}{\hbar^2} V(\vec{r}') \psi(\vec{r}') \right) d^3 r'$$

$$\Rightarrow \psi(\vec{r}) = e^{i \vec{k} \cdot \vec{r}} + \frac{2m}{\hbar^2} \int d^3 r' \left[ \frac{-e^{i k |\vec{r} - \vec{r}'|}}{4\pi |\vec{r} - \vec{r}'|} \right] V(\vec{r}') \psi(\vec{r}')$$

Ans 9) 
$$f_B(\theta) = \frac{-m}{2\pi\hbar^2} \int d^3r' e^{i\vec{q}\cdot\vec{r}'} V(\vec{r}')$$

for spherically symmetric potentials :

$$f_B(\theta) = \frac{-2m}{\hbar^2} \int_0^\infty r V(r) \sin(qr) dr$$

$$q = |\vec{k}_s - \vec{k}| = 2k \sin(\theta/2)$$

(a) (1)  $V(\vec{r}) = V_0 \lambda^3 \delta^3(\vec{r})$

$$\begin{aligned} f_B(\theta) &= \frac{-m}{2\pi\hbar^2} \int d^3r' e^{i\vec{q}\cdot\vec{r}'} [V_0 \lambda^3 \delta^3(\vec{r}')] \\ &= \frac{-m V_0 \lambda^3}{2\pi\hbar^2} \end{aligned}$$

(2)  $V(\vec{r}) = V_0 \lambda^5 \nabla^2 \delta^3(\vec{r})$

$$\int d^3r' e^{i\vec{q}\cdot\vec{r}'} \nabla^2 \delta^3(\vec{r}') = \int d^3r' \delta^3(\vec{r}') \nabla^2 (e^{i\vec{q}\cdot\vec{r}'})$$

$$\nabla^2 e^{i\vec{q}\cdot\vec{r}'} = (i\vec{q}) \cdot (i\vec{q}) e^{i\vec{q}\cdot\vec{r}'} = -q^2 e^{i\vec{q}\cdot\vec{r}'}$$

$$\int d^3r' \delta^3(\vec{r}') (-q^2 e^{i\vec{q}\cdot\vec{r}'}) = -q^2$$

$$f_B(\theta) = \frac{-m}{2\pi\hbar^2} (V_0 \lambda^5) (-q^2) = \frac{m V_0 \lambda^5 q^2}{2\pi\hbar^2}$$

$$(3) \int d^3r e^{i\vec{q}\cdot\vec{r}} e^{-\alpha r^2} = \left(\frac{\pi}{\alpha}\right)^{3/2} e^{-q^2/4\alpha}$$

where  $\alpha = 1/4\lambda^2$

$$\int d^3r e^{i\vec{q}\cdot\vec{r}} e^{-r^2/(4\lambda^2)} = (4\pi\lambda^2)^{3/2} e^{-q^2\lambda^2} = 8\pi \lambda^3 e^{-q^2\lambda^2}$$

$$f_B(\theta) = \frac{-m}{2\pi\hbar^2} V_0 (8\pi^{3/2} \lambda^3 e^{-q^2\lambda^2}) = \frac{-4mV_0 \lambda^3 \sqrt{\pi}}{\hbar^2} e^{-q^2\lambda^2}$$

$$(b) (i) f_B(\theta) = \frac{-2mV_0}{\hbar^2 q} \int_0^R r \sin(qr) dr$$

integrate by parts

$$\int_0^R r \sin(qr) dr = \frac{\sin(qR) - qR \cos(qR)}{q^2}$$

$$\Rightarrow f_B(\theta) = \frac{-2mV_0}{\hbar^2 q} \left[ \frac{\sin qR - qR \cos qR}{q^2} \right]$$

for  $kR = 1/2$ ,  $q = 2k \sin(\theta/2)$   
 $= \sin(\theta/2)$

zeros of  $f_B(\theta)$ :  $\tan qR = qR$

happens when  $qR = 0, 4.49, \dots$   
 but  $1 < 4.49$ ,  
 no valid scattering  $\theta$  is possible

$$(ii) \sigma_T = \int |f_B(\theta)|^2 d\Omega = \left( \frac{2mV_0 R^3}{\hbar^2} \right)^2 \frac{2\pi}{k^2 R^2} I$$

$$I = \int_0^d \frac{(\sin x - x \cos x)^2}{x^5} x dx$$

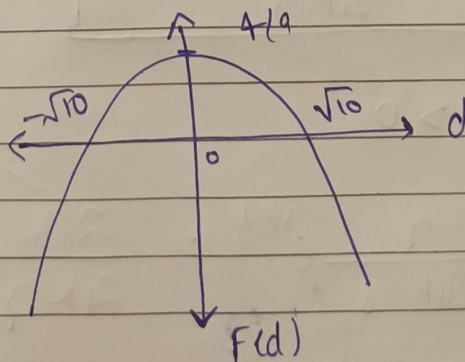
$$\Rightarrow \sigma_T = \pi R^2 \left[ \frac{2mV_0 R^2}{\hbar^2} \right]^2 \frac{8}{d^2} \int_0^d \frac{(\sin x - x \cos x)^2}{x^5} dx$$

$$\Rightarrow \sigma_T = \pi R^2 \left[ \frac{2mV_0 R^2}{\hbar^2} \right]^2 \frac{(2d^4 - 2d^2 + 2d \sin 2d - 1 + \cos 2d)}{d^6}$$

(iii) ~~as  $d \rightarrow 0$ : F~~

$$\Rightarrow \sigma_T = \pi R^2 \left[ \frac{2mV_0 R^2}{\hbar^2} \right]^2 F(d)$$

$$(iii) \text{ as } d \rightarrow 0, F(d) \rightarrow \frac{4}{9} - \frac{2d^2}{45} + \dots$$



$$(iv) F(d)_{\max} = 4/9$$

$$\sigma_T^{\max} = \pi R^2 \left( \frac{2mV_0 R^2}{\hbar^2} \right)^2 \frac{4}{9} = \boxed{\pi R^2 \left( \frac{4mV_0 R^2}{3\hbar^2} \right)^2}$$

Ans 10)  $\psi(x) = \phi_0(x) + \int_{-\infty}^{\infty} G(x, x') U(x') \psi(x') dx'$

$$V(x) = \frac{\hbar^2}{2m} \gamma \delta(x) \quad , \quad U(x) = \gamma \delta(x)$$

$$\Rightarrow \psi(x) = e^{ikx} + \frac{\gamma}{2ik} e^{i|k|x} \psi(0)$$

$$\psi(0) = e^0 + \frac{\gamma}{2ik} \psi(0) \quad \Rightarrow \quad \cancel{\psi(0)} = 1 + \frac{\gamma}{2ik} \cancel{\psi(0)}$$

$$\Rightarrow \boxed{\psi(0) = \frac{2ik}{2ik - \gamma}}$$

$$\psi(x) = e^{ikx} + \left( \frac{\gamma}{2ik} \right) \left( \frac{2ik}{2ik - \gamma} \right) e^{i|k|x}$$

$$= \boxed{e^{ikx} + \frac{\gamma}{2ik - \gamma} e^{i|k|x}}$$

Now for  $x > 0$

$$\psi(x) = e^{ikx} + \frac{\gamma}{2ik - \gamma} e^{ikx} = \frac{2ik}{2ik - \gamma} e^{ikx}$$

$$\Rightarrow \boxed{t = \frac{2ik}{2ik - \gamma}}$$

For  $x < 0$ ,  $|x| = -x$

$$\Rightarrow \psi(x) = e^{ikx} + \frac{\gamma}{2ik - \gamma} e^{-ikx} \Rightarrow \boxed{r = \frac{\gamma}{2ik - \gamma}}$$

$$T = \frac{4k^2}{4k^2 + \gamma^2}$$

$$R = \frac{\gamma^2}{4k^2 + \gamma^2}$$

$$\text{Ans 11)} \quad T_{fi} = \frac{2\pi}{\hbar} \int_{p_f=0}^{\infty} |V_{fi}|^2 \delta\left(\frac{p_f^2}{2m} - \frac{p_i^2}{2m}\right) \frac{L^3 p_f^2 dp_f d\Omega}{(2\pi\hbar)^3}$$

$$\text{let } E_f = \frac{p_f^2}{2m} \Rightarrow dE_f = \frac{p_f}{m} dp_f$$

$$\Rightarrow T_{fi} = \frac{2\pi}{\hbar} \int |V_{fi}|^2 \delta(E_f - E_i) \frac{L^3 p_f^2 d\Omega}{(2\pi\hbar)^3} \frac{dp_f}{p_f} dE_f$$

$$\Rightarrow \frac{dT_{fi}}{d\Omega} = \frac{2\pi}{\hbar} |V_{fi}|^2 \frac{L^3 m p_i}{(2\pi\hbar)^3}$$

$$V_{fi} = \langle f | V(r) | i \rangle = \frac{1}{L^3} \int d^3r e^{i(\mathbf{k}_i - \mathbf{k}_f) \cdot \mathbf{r}} V(r)$$

$$f_B(0) = \frac{-m}{2\pi\hbar^2} \int d^3r e^{i(\mathbf{k}_i - \mathbf{k}_f) \cdot \mathbf{r}} V(r)$$

$$\Rightarrow V_{fi} = \frac{1}{L^3} \left( \frac{-2\pi\hbar^2}{m} f_B(0) \right)$$

$$\Rightarrow |V_{fi}|^2 = \frac{1}{L^6} \frac{4\pi^2 \hbar^4}{m^2} |f_B(0)|^2$$

substituting into integral :

$$\frac{dT_f}{d\Omega} = \frac{2\pi}{\hbar} \left[ \frac{1}{L^3} \frac{4\pi^2 \hbar^4}{m^2} |f_B(\theta)|^2 \right] \frac{L^3 m P_i}{(2\pi\hbar)^3}$$

$$\Rightarrow \frac{dT_f}{d\Omega} = \frac{P_i}{mL^3} |f_B(\theta)|^2$$

$$j_{in} = P_i / mL^3, \quad P_i = P_f \text{ for elastic scattering}$$

$$\Rightarrow \boxed{\frac{dT_f}{d\Omega} = j_{in} |f_B(\theta)|^2}$$

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{1}{j_{in}} \frac{dT_f}{d\Omega} = |f_B(\theta)|^2}$$

b

# HOMEWORK 6

Page No. ....

Ans 1) a)  $\frac{1}{c^2} \frac{\partial^2 \psi(\vec{r}, t)}{\partial t^2} = \nabla^2 \psi(\vec{r}, t) - \left(\frac{m_e c}{\hbar}\right)^2 \psi(\vec{r}, t)$

$$\Rightarrow \frac{1}{c^2} \psi^* \frac{\partial^2 \psi(\vec{r}, t)}{\partial t^2} = \psi^* \nabla^2 \psi(\vec{r}, t) - \left(\frac{m_e c}{\hbar}\right)^2 \psi^* \psi$$

also  $\frac{1}{c^2} \frac{\partial^2 \psi^*}{\partial t^2} = \nabla^2 \psi^* - \left(\frac{m_e c}{\hbar}\right)^2 \psi^*$

$$\Rightarrow \frac{1}{c^2} \psi \frac{\partial^2 \psi^*}{\partial t^2} = \psi \nabla^2 \psi^* - \left(\frac{m_e c}{\hbar}\right)^2 \psi \psi^*$$

$$\Rightarrow \frac{1}{c^2} \left( \psi^* \frac{\partial^2 \psi}{\partial t^2} - \psi \frac{\partial^2 \psi^*}{\partial t^2} \right) = (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*)$$

$$\Rightarrow \frac{1}{c^2} \frac{\partial}{\partial t} \left( \psi^* \frac{\partial \psi}{\partial t} - \psi \frac{\partial \psi^*}{\partial t} \right)$$

$$= \vec{\nabla} \cdot (\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*)$$

$$\Rightarrow \frac{i\hbar}{2m_e c^2} \frac{\partial}{\partial t} \left( \psi^* \frac{\partial \psi}{\partial t} - \psi \frac{\partial \psi^*}{\partial t} \right) + \vec{\nabla} \cdot \left[ \frac{-i\hbar}{2m_e} (\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*) \right] = 0$$

$$\rho(\vec{r}, t) = \frac{i\hbar}{2m_e c^2} \left[ \psi^* \frac{\partial \psi}{\partial t} - \psi \frac{\partial \psi^*}{\partial t} \right]$$

$$\vec{J}(\vec{r}, t) = \frac{\hbar}{2im_e} [\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*]$$

$$\Rightarrow \boxed{\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0}$$

b)  $\Psi(\vec{r}, t) = A \exp\{i(\vec{p} \cdot \vec{r} - Et)/\hbar\}$

$$\frac{\partial \Psi}{\partial t} = \frac{-iE}{\hbar} \Psi \quad \text{and} \quad \frac{\partial \Psi^*}{\partial t} = \frac{iE}{\hbar} \Psi^*$$

$$P(\vec{r}, t) = \frac{i\hbar}{2m_e c^2} \left[ \Psi^* \left( \frac{-iE}{\hbar} \Psi \right) - \Psi \left( \frac{iE}{\hbar} \Psi^* \right) \right]$$

$$P(\vec{r}, t) = \frac{i\hbar}{2m_e c^2} \left( \frac{-2iE}{\hbar} |\Psi|^2 \right) = \frac{E}{m_e c^2} |\Psi|^2$$

For  $E < 0$ ,  $P < 0$  as  $|\Psi|^2 > 0$

but when  $\Psi(\vec{r}, t) \in \mathbb{R} \Rightarrow \Psi^* = \Psi$

$$P(\vec{r}, t) = \frac{i\hbar}{2m_e c^2} \left[ \Psi \frac{\partial \Psi}{\partial t} - \Psi \frac{\partial \Psi}{\partial t} \right] = 0$$

c) No, we can not ~~not~~ define a standard single particle directly from a single component K-G eq<sup>n</sup>.

Ans2) a)  $\alpha_j = \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix}$ ,  $\beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$

(a)  $\alpha_j^2 = I_4$ ,  $\beta^2 = I_4$

$$\alpha_j^2 = \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix} = \begin{pmatrix} \sigma_j^2 & 0 \\ 0 & \sigma_j^2 \end{pmatrix} \Rightarrow \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}$$

$$\beta^2 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} = I_4$$

b)  $\{d_i, d_j\}$

$$d_i d_j = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix} = \begin{pmatrix} \sigma_i \sigma_j & 0 \\ 0 & \sigma_i \sigma_j \end{pmatrix}$$

$$\{d_i, d_j\} = \begin{pmatrix} \{\sigma_i, \sigma_j\} & 0 \\ 0 & \{\sigma_i, \sigma_j\} \end{pmatrix} = \begin{pmatrix} 2\delta_{ij} I & 0 \\ 0 & 2\delta_{ij} I \end{pmatrix}$$

$$= \boxed{2\delta_{ij} I_4}$$

$$\{d_i, \beta\}: d_i \beta = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} = \begin{pmatrix} 0 & -\sigma_i \\ \sigma_i & 0 \end{pmatrix}$$

$$\beta d_i = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}$$

$$\{d_i, \beta\} = \begin{pmatrix} 0 & -\sigma_i + \sigma_i \\ \sigma_i - \sigma_i & 0 \end{pmatrix} = \boxed{0}$$

c)  $d_j d_k = i\epsilon_{jkl} \begin{pmatrix} \sigma_l & 0 \\ 0 & \sigma_l \end{pmatrix}$

$$d_j d_k = \begin{pmatrix} \sigma_j \sigma_k & 0 \\ 0 & \sigma_j \sigma_k \end{pmatrix}, \quad \sigma_j \sigma_k = \delta_{jk} I + i\epsilon_{jkl} \sigma_l$$

$$d_j d_k = \begin{pmatrix} \delta_{jk} I + i\epsilon_{jkl} \sigma_l & 0 \\ 0 & \delta_{jk} I + i\epsilon_{jkl} \sigma_l \end{pmatrix}$$

$$d_j d_k = \delta_{jk} I_4 + i\epsilon_{jkl} \begin{pmatrix} \sigma_l & 0 \\ 0 & \sigma_l \end{pmatrix}$$

b)  $\{d_i, d_j\}$ 

$$d_i d_j = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix} = \begin{pmatrix} \sigma_i \sigma_j & 0 \\ 0 & \sigma_i \sigma_j \end{pmatrix}$$

$$\{d_i, d_j\} = \begin{pmatrix} \{\sigma_i, \sigma_j\} & 0 \\ 0 & \{\sigma_i, \sigma_j\} \end{pmatrix} = \begin{pmatrix} 2\delta_{ij} I & 0 \\ 0 & 2\delta_{ij} I \end{pmatrix}$$

$$= \boxed{2\delta_{ij} I_4}$$

$$\{d_i, \beta\}: \quad d_i \beta = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} = \begin{pmatrix} 0 & -\sigma_i \\ \sigma_i & 0 \end{pmatrix}$$

$$\beta d_i = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}$$

$$\{d_i, \beta\} = \begin{pmatrix} 0 & -\sigma_i + \sigma_i \\ \sigma_i - \sigma_i & 0 \end{pmatrix} = \boxed{0}$$

$$c) \quad d_j d_k = i\epsilon_{jkl} \begin{pmatrix} \sigma_l & 0 \\ 0 & \sigma_l \end{pmatrix}$$

$$d_j d_k = \begin{pmatrix} \sigma_j \sigma_k & 0 \\ 0 & \sigma_j \sigma_k \end{pmatrix}, \quad \sigma_j \sigma_k = \delta_{jk} I + i\epsilon_{jkl} \sigma_l$$

$$d_j d_k = \begin{pmatrix} \delta_{jk} I + i\epsilon_{jkl} \sigma_l & 0 \\ 0 & \delta_{jk} I + i\epsilon_{jkl} \sigma_l \end{pmatrix}$$

$$d_j d_k = \delta_{jk} I_4 + i\epsilon_{jkl} \begin{pmatrix} \sigma_l & 0 \\ 0 & \sigma_l \end{pmatrix}$$

$$= i\epsilon_{jkl} \begin{pmatrix} \sigma_l & 0 \\ 0 & \sigma_l \end{pmatrix}$$

Ans 3)  $i\hbar \frac{\partial \Psi(\vec{r}, t)}{\partial t} = H \Psi(\vec{r}, t)$

$$H = c\vec{\alpha} \cdot \vec{p} + \beta m_e c^2$$

(a)  $\alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$

$$\vec{p} = -i\hbar \nabla = (-i\hbar \partial_x, -i\hbar \partial_y, -i\hbar \partial_z)$$

$$\vec{\sigma} \cdot \vec{p} = \alpha_x p_x + \alpha_y p_y + \alpha_z p_z = \begin{pmatrix} 0 & \vec{\sigma} \cdot \vec{p} \\ \vec{\sigma} \cdot \vec{p} & 0 \end{pmatrix}$$

$$\vec{\sigma} \cdot \vec{p} = \begin{bmatrix} p_z & p_x - ip_y \\ p_x + ip_y & -p_z \end{bmatrix} \quad (c)$$

$$H = \begin{bmatrix} m_e c^2 & 0 & c p_z & c(p_x + ip_y) \\ 0 & m_e c^2 & c(p_x + ip_y) & -c p_z \\ c p_z & c(p_x + ip_y) & -m_e c^2 & 0 \\ c(p_x + ip_y) & -c p_z & 0 & -m_e c^2 \end{bmatrix}$$

$$i\hbar \frac{\partial}{\partial t} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} = \begin{pmatrix} m_e c^2 & 0 & -i\hbar c \partial_z & -i\hbar c (\partial_x - i\partial_y) \\ 0 & m_e c^2 & -i\hbar c (\partial_x + i\partial_y) & i\hbar c \partial_z \\ -i\hbar c \partial_z & -i\hbar c (\partial_x - i\partial_y) & -m_e c^2 & 0 \\ -i\hbar c (\partial_x + i\partial_y) & i\hbar c \partial_z & 0 & -m_e c^2 \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}$$

(b)  ~~$\hat{v} = c\vec{\alpha}$~~ ,  $\hat{v} = \frac{dr}{dt} = \frac{1}{i\hbar} [\vec{r}, H]$

$$\hat{v} = \frac{1}{i\hbar} [\vec{r}, c\vec{\alpha} \cdot \vec{p} + \beta m_e c^2]$$

$$[x_i, p_j] = i\hbar \delta_{ij}$$

$$\hat{v}_i = \frac{1}{i\hbar} \left[ x_i, \sum_j c \alpha_j p_j \right] = \frac{c}{i\hbar} \sum_j \alpha_j [x_i, p_j]$$

$$\Rightarrow \hat{v}_i = \frac{c}{i\hbar} \sum_j \alpha_j (i\hbar \delta_{ij}) = c \alpha_i$$

$$\Rightarrow \boxed{\hat{v} = c \hat{\alpha}}$$

$$c) \quad i\hbar \frac{\partial \Psi}{\partial t} = -i\hbar c \bar{\alpha} \cdot \nabla \Psi + \beta m_e c^2 \Psi \quad (1)$$

$$-i\hbar \frac{\partial \Psi^\dagger}{\partial t} = i\hbar c (\nabla \Psi^\dagger) \cdot \bar{\alpha} + m_e c^2 \Psi^\dagger \beta \quad (2)$$

$$\Psi^\dagger \times (1) - (2) \times \Psi$$

$$\Rightarrow i\hbar \left( \Psi^\dagger \frac{\partial \Psi}{\partial t} - \frac{\partial \Psi^\dagger}{\partial t} \Psi \right) = -i\hbar c \left[ \Psi^\dagger \bar{\alpha} \cdot \nabla \Psi + (\nabla \Psi^\dagger) \cdot \bar{\alpha} \Psi \right]$$

$$\Rightarrow \boxed{i\hbar \frac{\partial}{\partial t} (\Psi^\dagger \Psi) = -i\hbar c \nabla \cdot (\Psi^\dagger \bar{\alpha} \Psi)}$$

$$\Rightarrow \boxed{\frac{\partial}{\partial t} (\Psi^\dagger \Psi) + \nabla \cdot (c \Psi^\dagger \bar{\alpha} \Psi) = 0}$$

$$\Rightarrow \boxed{\rho + \nabla \cdot \vec{J} = 0}$$

Ans 4) (a)  $[\vec{L}, H] = i\hbar c (\vec{\alpha} \times \vec{p})$

$$\vec{L} = \vec{r} \times \vec{p}, \quad L_i = \epsilon_{ijk} r_j p_k$$

$$[L_i, H] = [\epsilon_{ijk} r_j p_k, c\vec{\alpha} \cdot \vec{p} + \beta m_e c^2]$$

$$= c \sum_m \alpha_m [\epsilon_{ijk} r_j p_k, p_m]$$

$$= c \epsilon_{ijk} \sum_m \alpha_m [\sigma_j, p_m] p_k$$

$$[\sigma_j, p_m] = i\hbar \delta_{jm}$$

$$[L_i, H] = c \epsilon_{ijk} \sum_m \alpha_m (i\hbar \delta_{jm}) p_k = i\hbar c \sum_k \epsilon_{ikj} \alpha_j p_k$$

$$= i\hbar c (\vec{\alpha} \times \vec{p})_i$$

$$\Rightarrow [L, H] = i\hbar c (\vec{\alpha} \times \vec{p}) \neq 0$$

↓

angular momentum not conserved

(b)  $\Sigma = \begin{pmatrix} \sigma & 0 \\ 0 & \sigma \end{pmatrix} \quad \Sigma_x = -i\alpha_1 \alpha_2$

$$\Sigma_x = -i \begin{pmatrix} 0 & \sigma_y \\ \sigma_y & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma_z \\ \sigma_z & 0 \end{pmatrix} = -i \begin{pmatrix} \sigma_y \sigma_z & 0 \\ 0 & \sigma_y \sigma_z \end{pmatrix}$$

$$\sigma_j \sigma_k = \delta_{jk} I + i \epsilon_{jkl} \sigma_l, \quad \sigma_y \sigma_z = i \sigma_x$$

$$\Sigma_x = \begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}$$

(c)  $S = (\hbar/2) \Sigma$

$$\Rightarrow S^2 = S_x^2 + S_y^2 + S_z^2 = \frac{\hbar^2}{4} (\Sigma_x^2 + \Sigma_y^2 + \Sigma_z^2)$$

$$= \frac{3\hbar^2}{4} I_4$$

$\Sigma_z = \begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}$  has eigenvalues  $\pm 1$

$\Rightarrow$  eigenvalues of  $S$  are  $\pm \hbar/2$

(d)  $\bar{J} = \bar{L} + \bar{S}$

$$[S_i, H] = \frac{\hbar}{2} \left[ \Sigma_i, c \sum_j \alpha_j p_j \right] = \frac{\hbar c}{2} \sum_j [\Sigma_i, \alpha_j] p_j$$

evaluating  $[\Sigma_i, \alpha_j]$ :

$$\begin{pmatrix} \sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix} \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix} - \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix} \begin{pmatrix} \sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix} = \begin{pmatrix} 0 & [\sigma_i, \sigma_j] \\ [\sigma_i, \sigma_j] & 0 \end{pmatrix}$$

$$\Rightarrow [\Sigma_i, \alpha_j] = 2i \epsilon_{ijk} \alpha_k$$

$$[S_i, H] = -i\hbar c (\bar{\alpha} \times \bar{p})_i$$

$$\Rightarrow [S, H] = -i\hbar c [\bar{\alpha} \times \bar{p}]$$

$$[\bar{J}, H] = [\bar{L}, H] + [\bar{S}, H] = i\hbar c (\bar{\alpha} \times \bar{p}) - i\hbar c (\bar{\alpha} \times \bar{p})$$

$$= 0$$

$$(e) [\bar{H}, \bar{S} \cdot \bar{p}] = \sum_i [\bar{H}, S_i p_i] = \sum_i ([\bar{H}, S_i] p_i + S_i [\bar{H}, p_i])$$

$$\text{now } [\bar{H}, p_i] = 0$$

$$[\bar{H}, \bar{S} \cdot \bar{p}] = \sum_i [\bar{H}, S_i] p_i = - \sum_i [\bar{H}, S_i] p_i$$

$$\Rightarrow [\bar{H}, \bar{S} \cdot \bar{p}] = i\hbar c (\bar{\sigma} \times \bar{p}) \cdot \bar{p} = 0$$

$$\Rightarrow \boxed{[\bar{H}, \bar{S} \cdot \bar{p}] = 0}$$

$$(f) \hat{n} = \frac{\bar{\Sigma} \cdot \hat{p}}{|\bar{p}|}, \quad \hat{n}^2 = \frac{(\bar{\Sigma} \cdot \hat{p})^2}{|\bar{p}|^2}$$

$$(\bar{\Sigma} \cdot \bar{A})(\bar{\Sigma} \cdot \bar{B}) = \bar{A} \cdot \bar{B} I + i \bar{\Sigma} \cdot (\bar{A} \times \bar{B}), \quad \bar{A} = \bar{B} = \bar{p}$$

$$(\bar{\Sigma} \cdot \bar{p})^2 = |\bar{p}|^2 I + i \bar{\Sigma} \cdot (\bar{p} \times \bar{p}) = |\bar{p}|^2 I$$

$$\hat{n}^2 = \frac{|\bar{p}|^2}{|\bar{p}|^2} = I_4$$

its eigenvalues are  $\pm 1$

$$\psi_+ = \begin{pmatrix} \chi_+ \\ c_1 \chi_+ \end{pmatrix}$$

$$\psi_- = \begin{pmatrix} \chi_- \\ c_2 \chi_- \end{pmatrix}$$

such that :

$$(\bar{\sigma} \cdot \hat{n}) \chi_{\pm} = \pm \chi_{\pm}$$

Ans 5 (a)  $i\hbar \frac{\partial \Psi(\vec{r}, t)}{\partial t} = H \Psi(\vec{r}, t) = (c \vec{\alpha} \cdot \vec{p} + \beta m_e c^2) \Psi$

$$E \Psi = (c \vec{\alpha} \cdot \vec{p} + \beta m_e c^2) \Psi$$

$$\Psi = \begin{pmatrix} u \\ w \end{pmatrix} \quad \vec{\alpha} = \begin{pmatrix} 0 & \vec{\sigma} \\ \vec{\sigma} & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$$

$$E \begin{pmatrix} u \\ w \end{pmatrix} = \begin{pmatrix} m_e c^2 & c \vec{\sigma} \cdot \vec{p} \\ c \vec{\sigma} \cdot \vec{p} & -m_e c^2 \end{pmatrix} \begin{pmatrix} u \\ w \end{pmatrix}$$

This gives :

$$\begin{aligned} (E - m_e c^2) u - c (\vec{\sigma} \cdot \vec{p}) w &= 0 \\ -c (\vec{\sigma} \cdot \vec{p}) u + (E + m_e c^2) w &= 0 \end{aligned}$$

(b)  $w = \frac{c (\vec{\sigma} \cdot \vec{p})}{E + m_e c^2} u$

$$\Rightarrow (E - m_e c^2) u - c (\vec{\sigma} \cdot \vec{p}) \left[ \frac{c (\vec{\sigma} \cdot \vec{p})}{E + m_e c^2} u \right] = 0$$

$$\Rightarrow (E - m_e c^2) (E + m_e c^2) u - c^2 (\vec{\sigma} \cdot \vec{p})^2 u = 0$$

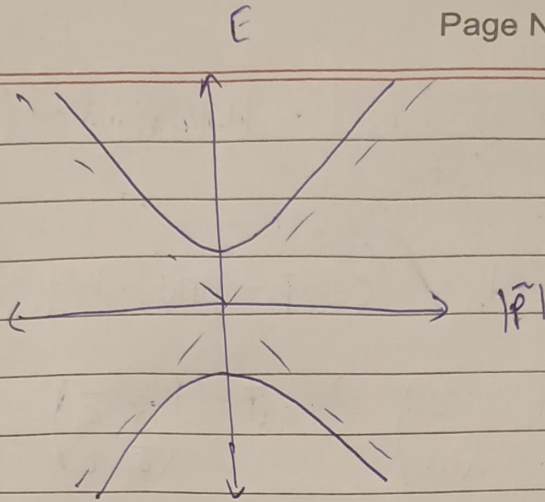
identity :  $(\vec{\sigma} \cdot \vec{A})(\vec{\sigma} \cdot \vec{B}) = \vec{A} \cdot \vec{B} + i \vec{\sigma} \cdot (\vec{A} \times \vec{B})$

$$\Rightarrow (\vec{\sigma} \cdot \vec{p})^2 = |\vec{p}|^2 = p^2 \quad \vec{A} = \vec{B} = \vec{p}$$

$$\Rightarrow (E^2 - m_e^2 c^4 - p^2 c^2) u = 0$$

$$E^2 = p^2 c^2 + m_e^2 c^4 \Rightarrow E(p) = \pm \sqrt{p^2 c^2 + m_e^2 c^4}$$

$$\frac{E}{m_e c^2} = \pm \sqrt{1 + |\vec{p}|^2}$$



$$(c) \quad E_{\pm} = \pm \sqrt{p^2 c^2 + m_e^2 c^4}$$

choose upper spinor  $\psi$

$$\omega = \frac{c(\vec{\sigma} \cdot \vec{p})}{E_{\pm} + m_e c^2} \psi$$

$$\psi^{(1)} = N \begin{pmatrix} 1 \\ 0 \\ \frac{cp_z}{E_{+} + m_e c^2} \\ \frac{c(p_x + ip_y)}{E_{+} + m_e c^2} \end{pmatrix}$$

$$\psi^{(2)} = N \begin{pmatrix} 0 \\ 1 \\ \frac{c(p_x - ip_y)}{E_{+} + m_e c^2} \\ -\frac{cp_z}{E_{+} + m_e c^2} \end{pmatrix}$$

For  $E_{-} = -\sqrt{p^2 c^2 + m_e^2 c^4}$ :

$$\psi^{(3)} = \begin{pmatrix} \frac{-cp_z}{E_{-} + m_e c^2} \\ \frac{-c(p_x + ip_y)}{E_{-} + m_e c^2} \\ 1 \\ 0 \end{pmatrix}$$

$$\psi^{(4)} = \begin{pmatrix} \frac{-c(p_x - ip_y)}{E_{-} + m_e c^2} \\ \frac{cp_z}{E_{-} + m_e c^2} \\ 0 \\ 1 \end{pmatrix}$$

(d)  $\psi^\dagger \psi = 1$  for normalization

$$1 = |N|^2 \left[ 1 + \frac{p^2 c^2}{(E + m_e c^2)^2} \right]$$

$$\Rightarrow |N|^2 \left[ 1 + \frac{E - m_e c^2}{E + m_e c^2} \right] \Rightarrow N = \sqrt{\frac{E + m_e c^2}{2E}}$$

$$\langle v \rangle = c \bar{\alpha}, \quad \psi = \begin{pmatrix} v \\ w \end{pmatrix}$$

$$\langle v \rangle = \psi^\dagger (c \bar{\alpha}) \psi = c (v^\dagger w^\dagger) \begin{pmatrix} 0 & \sigma \\ \sigma & 0 \end{pmatrix} \begin{pmatrix} v \\ w \end{pmatrix}$$

$$\langle v \rangle = c (v^\dagger \bar{\sigma} w + w^\dagger \bar{\sigma} v)$$

$$\Rightarrow \boxed{\langle v \rangle = \bar{p} c^2 / E}$$

$$\langle S \rangle = \psi^\dagger \frac{\hbar}{2} \begin{pmatrix} \bar{\sigma} & 0 \\ 0 & \bar{\sigma} \end{pmatrix} \psi = \boxed{\frac{\hbar}{2} N^2 (v^\dagger \bar{\sigma} v + w^\dagger \bar{\sigma} w)}$$

taking  $\psi^{(1)}$ ,  $\bar{p} = p_z \hat{k}$ ,  $\langle S \rangle = (0, 0, \hbar/2)$

Ans 56) a)  $\begin{pmatrix} m_e c^2 + V & c \bar{\sigma} \cdot \bar{p} \\ c \bar{\sigma} \cdot \bar{p} & -m_e c^2 + V \end{pmatrix} \begin{pmatrix} \psi \\ \chi \end{pmatrix} = E \begin{pmatrix} \psi \\ \chi \end{pmatrix}$

$$E = m_e c^2 + E$$

$$c(\bar{\sigma} \cdot \bar{p}) \psi + (-2m_e c^2 + V - E) \chi = 0$$

$$\Rightarrow \chi = \frac{c(\bar{\sigma} \cdot \bar{p})}{2m_e c^2 + E - V} \psi$$

For  $E, V \ll 2m_e c^2$ :

$$\chi \approx \frac{1}{2m_e c} \left\{ 1 - \frac{E-V}{2m_e c^2} \right\} (\vec{\sigma} \cdot \vec{p}) \psi$$

Substitute  $\chi$  into top row eq<sup>n</sup>.

$$E \psi = V \psi + c (\vec{\sigma} \cdot \vec{p}) \left[ \frac{1}{2m_e c} \left( 1 - \frac{E-V}{2m_e c^2} \right) (\vec{\sigma} \cdot \vec{p}) \right] \psi$$

$$E \psi = \left[ V + \frac{(\vec{\sigma} \cdot \vec{p})^2}{2m_e} - \frac{1}{4m_e^2 c^2} (\vec{\sigma} \cdot \vec{p})(E-V)(\vec{\sigma} \cdot \vec{p}) \right] \psi$$

$$(\vec{\sigma} \cdot \vec{p})^2 = p^2$$

$$\Rightarrow \text{first two terms give: } \left( \frac{p^2}{2m_e} + V \right) \psi = E \psi$$

$$[\vec{p}, V] = -i\hbar \nabla V$$

(i) relativistic KE correction:

$$E-V \approx \frac{p^2}{2m_e}, \quad \text{gives } \frac{-p^4}{8m_e^3 c^2}$$

$$(ii) \text{ Spin-Orbit Coupling: } \frac{\hbar}{8m_e^2 c^2} \nabla^2 V$$

$$(iii) \text{ Darwin term: } \frac{\hbar^2}{8m_e^2 c^2} \nabla^2 V$$

(b) Zeeman effect

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

$$\vec{p} \rightarrow \vec{\pi} = \vec{p} + e\vec{A}$$

$$\begin{pmatrix} m_e c^2 & c \vec{\sigma} \cdot \vec{\pi} \\ c \vec{\sigma} \cdot \vec{\pi} & -m_e c^2 \end{pmatrix} \begin{pmatrix} \psi \\ \chi \end{pmatrix} = (m_e c^2 + e) \begin{pmatrix} \psi \\ \chi \end{pmatrix}$$

$$e \ll 2m_e c^2$$

$$c(\vec{\sigma} \cdot \vec{\pi})\psi - 2m_e c^2 \chi \approx 0 \Rightarrow \chi \approx \frac{\vec{\sigma} \cdot \vec{\pi}}{2m_e c} \psi$$

$$\Rightarrow e\psi = e(\vec{\sigma} \cdot \vec{\pi}) \left( \frac{\vec{\sigma} \cdot \vec{\pi}}{2m_e c} \right) \psi = \frac{(\vec{\sigma} \cdot \vec{\pi})^2}{2m_e} \psi$$

evaluate  $(\vec{\sigma} \cdot \vec{\pi})^2$  using  $(\vec{\sigma} \cdot \vec{U})(\vec{\sigma} \cdot \vec{V})$   
 $= \vec{U} \cdot \vec{V} + i\vec{\sigma} \cdot (\vec{U} \times \vec{V})$

$$(\vec{\sigma} \cdot \vec{\pi})^2 = \pi^2 + i\vec{\sigma} \cdot (\vec{\pi} \times \vec{\pi})$$

$$\vec{\pi} \times \vec{\pi} = e(\vec{p} \times \vec{A} + \vec{A} \times \vec{p})$$

$$= -ie\hbar \vec{B} \quad (\text{applying to last } \int^n)$$

$$\rightarrow (\vec{\sigma} \cdot \vec{\pi})^2 = (\vec{p} + e\vec{A})^2 + e\hbar \vec{\sigma} \cdot \vec{B}$$

$$e\psi = \left[ \frac{(\vec{p} + e\vec{A})^2}{2m_e} + \frac{e\hbar}{2m_e} \vec{\sigma} \cdot \vec{B} \right] \psi$$

$$\vec{S} = (\hbar/2)\vec{\sigma}, \quad g=2$$

$$\frac{e\hbar}{2m_e} \vec{\sigma} \cdot \vec{B} = \frac{e}{m_e} \left( \frac{\hbar \vec{\sigma}}{2} \right) \cdot \vec{B} = \frac{ge}{2m_e} \vec{S} \cdot \vec{B} \Rightarrow \left[ \frac{(\vec{p} + e\vec{A})^2}{2m_e} + g \frac{e}{2m_e} \vec{S} \cdot \vec{B} \right] \psi(x) = e\psi(x)$$